

40 A nucleus decays in a series of stages.

At each stage, an α -particle or β^- particle is emitted.

Which sequence of emitted particles forms a nucleus that is an isotope of the original nucleus?

A $\alpha, \alpha, \beta^-, \beta^-$

B β^-, β^-, α

C $\alpha, \alpha, \alpha, \alpha$

D $\beta^-, \beta^-, \beta^-$